ELEC2104/9704 Tutorial 4 Solutions (2025)

1 Question 1

A diode has $I_S = 10^{-16}$ A and $n = 1$.

(a) What is the diode current if the diode voltage is 0.675 V?

(b) What will be the diode voltage if the current increases by a factor of 3?

Solution:

For a diode, we have the equation below that describes the correlation between the diode voltage (V_D) and the diode current (I_D).

$$I_D = I_S \left[\exp\left(\frac{V_D}{nV_T} \right) - 1 \right] \tag{1}$$

Where:

- n is the ideality factor, which accounts for carrier recombination in the depletion region. For an ideal diode, $n = 1$, while real diodes typically have $1 < n < 2$. We almost always assume $n = 1$ in this course.
- I_S is the saturation current (also called reverse saturation current).
- $V_T = \frac{kT}{q}$ is the thermal voltage.

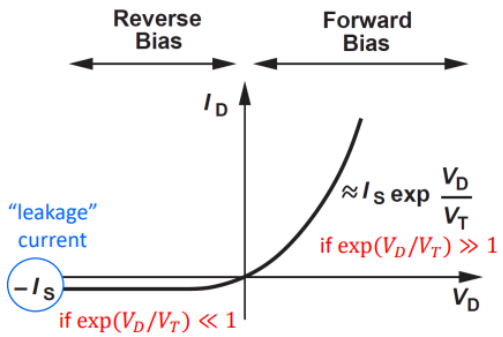


Figure 1: Diode I-V characteristic showing the exponential relationship in forward bias and the small leakage current ($-I_S$) in reverse bias. In forward bias where $V_D \gg V_T$, the exponential term dominates and $I_D \approx I_S e^{V_D/V_T}$.

1.1 Part A: What is the diode current if the diode voltage is 0.675 V?

The values for the ideality factor and saturation current are given in the question:

$$n = 1 \tag{2}$$

$$I_S = 10^{-16}\text{A} \tag{3}$$

Since no temperature is given, we need to assume room temperature. The thermal voltage is given by:

$$V_T = \frac{kT}{q} = \frac{(1.38 \times 10^{-23}\,\text{J/K}) \cdot T}{1.602 \times 10^{-19}\,\text{C}} \tag{4}$$

Note that the Boltzmann constant k and the elementary charge q are fundamental physical constants.

Room Temperature Conventions

Different sources use different room temperatures:

- $T = 293.15\,\text{K}$ (20°C): $V_T = 0.025\,\text{V} = 25\,\text{mV}$
- $T = 300\,\text{K}$ (26.85°C): $V_T = 0.026\,\text{V} = 26\,\text{mV}$

In exams, the temperature or thermal voltage should be provided. Note that it may be given implicitly in some other quantity. If neither is specified, you may use either 25 mV or 26 mV.

Here we'll use $T = 293.15\,\text{K}$:

$$V_T = 0.025\,\text{V} \tag{5}$$

Taking (3), (2), and (5) into (1), the diode current is determined:

$$I_D = I_S \left[\exp\left(\frac{V_D}{nV_T}\right) - 1\right] = (10^{-16}\,\text{A}) \left[\exp\left(\frac{0.675\,\text{V}}{0.025\,\text{V}}\right) - 1\right] = 5.32 \times 10^{-5}\,\text{A} = 53.2\,\mu\text{A} \tag{6}$$

If we had used $T = 300\,\text{K}$ ($V_T = 0.026\,\text{V}$), the result would be:

$$I_D \approx 1.8 \times 10^{-5}\,\text{A} = 18.83\,\mu\text{A} \tag{7}$$

1.2 Part B: What will be the diode voltage if the current increases by a factor of 3?

If the current increases by a factor of 3, the new current is:

$$I_{D,\text{new}} = 3I_D = 3 \cdot (5.32 \times 10^{-5}\,\text{A}) = 1.60 \times 10^{-4}\,\text{A} \tag{8}$$

From (1), we can rearrange the equation to derive an expression about the new diode voltage (try to memorise this equation for V_D in terms of I_D as it is important):

$$V_{D,\text{new}} = nV_T \ln\left(\frac{I_{D,\text{new}}}{I_S} + 1\right) \tag{9}$$

Taking (3), (2), (5) and (8) into (9), we can solve the new diode voltage.

$$V_{D,\text{new}} = V_T \ln\left(\frac{I_{D,\text{new}}}{I_S} + 1\right) = (0.025)\,\ln\left(\frac{1.60 \times 10^{-4}}{10^{-16}} + 1\right) = 0.702\,\text{V} \tag{10}$$

Note that above we used the $V_T = 25\,\text{mV}$ value for thermal voltage. For $V_T = 26\,\text{mV}$:

$$V_{D,\text{new}} = 0.731\,\text{V} \tag{11}$$

Note that you may have encountered rounding errors on this question. To reduce rounding errors, use the memory function in your calculator to save intermediate results.

2 Question 2

A diode is operating with $i_D = 2\,\text{mA}$ and $v_D = 0.82\,\text{V}$.

(a) What is I_S if $n = 1$?

(b) What is the diode current for $v_D = -5\,\text{V}$?

Solution:

2.1 Part 1: What is I_S if n = 1?

For a diode, we have the equation below that describes the correlation between the diode voltage (V_D) and the diode current (I_D).

$$I_D = I_S \left[\exp\left(\frac{V_D}{nV_T}\right) - 1\right] \tag{1}$$

Note how that the ideality factor (n) is often omitted, in which case $n = 1$ is implied.

For the given set of diode voltage (V_D), diode current (I_D), ideality factor (n) and the assumption that the temperature is 20°C (which sets the thermal voltage V_T), we have the following:

$$V_D = 0.82\text{V} \tag{2}$$

$$I_D = 2\text{mA} \tag{3}$$

$$n = 1 \tag{4}$$

$$V_T = 0.025\text{V} \tag{5}$$

Rearranging (1), we can derive an expression for the saturation current (I_S), where:

$$I_S = \frac{I_D}{\exp\left(\frac{V_D}{nV_T}\right) - 1} \tag{6}$$

Taking (2), (3), (4), and (5) into (6), we can determine I_S.

$$I_S = \frac{I_D}{\exp\left(\frac{V_D}{nV_T}\right) - 1} = \frac{2 \times 10^{-3}\,\text{A}}{\exp\left(\frac{0.82}{1\times 0.025}\right) - 1} = 1.14 \times 10^{-17}\,\text{A} \tag{7}$$

If $V_T = 26\,\text{mV}$ is used, the answer is $I_S = 4.02 \times 10^{-17}\,\text{A}$.

2.2 Part 2: What is the diode current for v_D = -5 V?

A diode acts as a one-way valve for current – it allows current to flow easily in the forward direction (positive voltage) but blocks current in the reverse direction (negative voltage). When reverse biased ($V_D < 0$), only a tiny leakage current flows, approximately equal to $-I_S$. Let's verify this. Taking the known n, V_D, I_S into (1), the diode current is determined:

$$I'_D = I_S \left[\exp\left(\frac{V'_D}{nV_T}\right) - 1\right] = \left(1.14 \times 10^{-17}\right)\left[\exp\left(\frac{-5}{0.025}\right) - 1\right] = -1.14 \times 10^{-17}\,\text{A} \approx -I_S \tag{8}$$

This confirms that in reverse bias, the diode current saturates at $-I_S$ (the reverse saturation current), which is typically on the order of 10^{-17} to 10^{-15} A – negligible for many practical circuits.

Permittivity of Silicon $\varepsilon_s = \varepsilon_r \cdot \varepsilon_0$

The permittivity of a material is expressed as:

$$\varepsilon = \varepsilon_r \cdot \varepsilon_0$$

where:

- ε_r is the relative permittivity (dielectric constant). For silicon: $\varepsilon_r = 11.7$ (commonly used in lectures). Despite the *constant* in "dielectric constant", it can vary depending on material properties and more!
- ε_0 is the permittivity of free space $= 8.85 \times 10^{-14}$ F/cm

This gives us $\varepsilon_{\text{Si}} = \varepsilon_r \cdot \varepsilon_0 = 11.7 \cdot 8.85 \times 10^{-14} = 1.035 \times 10^{-12}$ F/cm

Note: The course table of constants uses $\varepsilon_r = 11.9$ for silicon, which would give $\varepsilon_{\text{Si}} = 1.053 \times 10^{-12}$ F/cm

3 Question 3

A square *pn* junction diode is 5 mm on a side. The *p*-type side has a doping concentration of $10^{19}/\text{cm}^3$ and the *n*-type side has a doping concentration of $10^{16}/\text{cm}^3$.

(a) What is the zero-bias capacitance of the diode?

(b) What is the capacitance at a reverse bias of 4 V?

Solution:

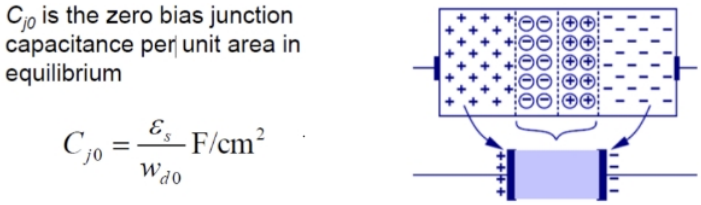


Figure 2: Junction capacitance in a pn diode. The depletion region acts as a parallel-plate capacitor with the neutral regions forming the plates. The capacitance per unit area C_{j0} depends on the permittivity ε_s and depletion width w_{d0}. The depletion width can be derived from the doping concentrations (N_A and N_D) and the built-in voltage V_{bi}.

Firstly, let's list the known parameters.

$$\text{Permittivity of Si:} \quad \varepsilon_{\text{Si}} = 11.7\,\varepsilon_0 = 11.7 \times 8.85 \times 10^{-14}\,\text{F/cm} \tag{1}$$

$$\text{Elementary charge:} \quad q = 1.602 \times 10^{-19}\,\text{C} \tag{2}$$

$$\text{Side length of the square junction:} \quad l = 5\,\text{mm} = 0.5\,\text{cm} \tag{3}$$

$$\text{Donor density:} \quad N_D = 10^{16}\,\text{cm}^{-3} \tag{4}$$

$$\text{Acceptor density:} \quad N_A = 10^{19}\,\text{cm}^{-3} \tag{5}$$

3.1 Part A: What is the zero-bias capacitance of the diode?

The *total* zero-bias capacitance of the diode has the formula below, where A is the junction area, C_{j0} is the junction capacitance *per unit area*, and W_{d0} is the zero-bias depletion width. Units are denoted

in square brackets for clarity.

$$C_0 \ [\mathrm{F}] = C_{j0}\,[\mathrm{F/cm^2}] \cdot A\,[\mathrm{cm^2}] = \left(\frac{\varepsilon_s}{W_{d0}}\right) \cdot A \ [\mathrm{F}] \tag{6}$$

Note again that here A is the area of the juncition cross-section and since the side length of the square junction is given, we can determine this area:

$$A = l \times l = (0.5\ \mathrm{cm})(0.5\ \mathrm{cm}) = 0.25\ \mathrm{cm}^2 \tag{7}$$

The depletion width has the formula below.

$$W_{d0} = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\phi_i} \tag{8}$$

It is good practice to mark which quantities are still unknown. Here, the built-in voltage ϕ_i (can also be denoted as V_{bi}) is unknown. Fortunately, we have sufficient parameters to determine ϕ_i using the formula presented in Lecture 4:

$$\phi_i = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) \tag{9}$$

Using assumptions for the thermal voltage (V_T) and the intrinsic carrier density of silicon (n_i), and the given values of doping concentrations from the question (N_A and N_D), the value of the built-in voltage is:

$$\phi_i = (0.025\ \mathrm{V}) \ln\left(\frac{(10^{19}\ \mathrm{cm^{-3}}) \cdot (10^{16}\ \mathrm{cm^{-3}})}{(10^{10}\ \mathrm{cm^{-3}})^{\,2}}\right) = 0.863\ \mathrm{V} \tag{10}$$

Or $\phi_i = 0.898\,\mathrm{V}$, if the thermal voltage is $V_T = 26\,\mathrm{mA}$.

Now, taking (10) into (8), the depletion width is determined.

$$W_{d0} = \sqrt{\frac{2\varepsilon_{\mathrm{Si}}}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\phi_i} \tag{11}$$

$$= \sqrt{\left(\frac{2 \cdot 1.035 \times 10^{-12}\ \mathrm{F/cm}}{1.602 \times 10^{-19}\ \mathrm{C}}\right)\left(\frac{1}{10^{19}\ \mathrm{cm^{-3}}} + \frac{1}{10^{16}\ \mathrm{cm^{-3}}}\right)(0.863\ \mathrm{V})} \tag{12}$$

$$W_{d0} = \sqrt{1.119 \times 10^{-9}\ \mathrm{cm^2}} = 3.34 \times 10^{-5}\ \mathrm{cm} = 0.334\ \mu\mathrm{m}. \tag{13}$$

Or $\phi_i = 0.341\,\mathrm{V}$, if the thermal voltage is $V_T = 26\,\mathrm{mA}$.

Dimensional Analysis Tip

When working with semiconductor equations, always perform *"unit algebra"* as a sanity check:

- Remember that $1\,\mathrm{F} = 1\,\mathrm{C/V}$ (Farads = Coulombs per Volt)
- Example, for the depletion width formula:

$$\sqrt{\frac{[\mathrm{F/cm}]}{[\mathrm{C}]} \cdot [\mathrm{cm^3}] \cdot [\mathrm{V}]} = \sqrt{\frac{[\mathrm{C/V}] \cdot [\mathrm{cm^2}] \cdot [\mathrm{V}]}{[\mathrm{C}]}} = \sqrt{[\mathrm{cm^2}]} = [\mathrm{cm}]$$

- This confirms our formula gives the correct units for distance!

Taking (1), (7), and (13) into (6), the zero-bias capacitance is determined:

$$C_{j0} = \frac{\varepsilon_{\mathrm{Si}}}{W_{d0}} = \frac{1.035 \times 10^{-12}\ \mathrm{F/cm}}{3.34 \times 10^{-5}\ \mathrm{cm}} = 3.0974 \times 10^{-8}\ \mathrm{F/cm^2} \tag{14}$$

The total capacitance across the junction cross-section area is then:

$$C_0 = C_{j0} \cdot A = \left(3.0974 \times 10^{-8}\ \text{F/cm}^2\right)(0.25\ \text{cm}^2) = 7.7435 \times 10^{-9}\ \text{F} = 7.74\ \text{nF}. \tag{15}$$

Or $C_{j0} = 3.0369 \times 10^{-8}\,\text{F/cm}^2$ and $C_0 = 7.59\,\text{nF}$ if the thermal voltage is $V_T = 26\,\text{mA}$.

Alternatively, we can directly use the formula for junction capacitance per unit area by substituting the depletion width expression:

$$C_{j0} = \sqrt{\frac{\varepsilon_s q}{2} \cdot \frac{N_A N_D}{N_A + N_D} \cdot \frac{1}{\phi_i}}\quad \text{F/cm}^2 \tag{16}$$

This combined formula is useful when you need to find capacitance directly from doping concentrations without explicitly calculating the depletion width first.

3.2 Part B: What is the capacitance at a reverse bias of 4 V?

For the pn junction with reverse bias V_R, the capacitance has the given formula, based on the zero-bias capacitance:

$$C' = \frac{C_0}{\sqrt{1 + \dfrac{V_R}{\phi_i}}} \tag{17}$$

Taking (10), (15) and $V_R = \mathbf{4\ V}$, the new capacitance with reverse bias is:

$$C' = \frac{C_0}{\sqrt{1 + \dfrac{V_R}{\phi_i}}} = \frac{7.7435\,\text{nF}}{\sqrt{1 + \dfrac{4\,V}{0.863\,V}}} = 3.263\ \text{nF} \tag{18}$$

Or $C' = 3.251\,\text{nF}$ if the thermal voltage is $V_T = 26\,\text{mA}$.

Common Assumptions in Semiconductor Problems

- **Thermal voltage:** $V_T = 25$ mV (at $T = 293.15$ K) or $V_T = 26$ mV (at $T = 300$ K)
- **Intrinsic carrier concentration (Si):**
  - Textbook approximation: $n_i = 10^{10}\ \text{cm}^{-3}$
  - Course table value: $n_i = 9.65 \times 10^9\ \text{cm}^{-3}$
- **Relative permittivity of Si:**
  - Lecture/textbook: $\varepsilon_r = 11.7$
  - Course table: $\varepsilon_r = 11.9$

In exams, use the values provided in the question, followed by the course table of constants (if not provided in the question).

4 Question 4

Consider a *pn* junction in forward bias and assume $V_T = 26\,\text{mV}$.

1. To obtain a current of 1 mA with a voltage of 750 mV, how should I_S be chosen?
2. If the diode cross-section area is now doubled, what voltage yields a current of 1 mA?

4.1 Part A: To obtain a current of 1 mA with a voltage of 750 mV, how should I_S be chosen?

For a diode, we have the equation below that describes the correlation between the diode voltage (V_D) and the diode current (I_D). Note that the ideality factor (n) is usually omitted when it equals one.

$$I_D = I_S \left[\exp\left(\frac{V_D}{V_T} \right) - 1 \right] \tag{1}$$

With the given diode voltage (V_D) and the diode current (I_D), we can simply solve (1) to get the required saturation current I_S :

$$I_S = \frac{I_D}{\exp\left(\frac{V_D}{V_T}\right) - 1} = \frac{1 \times 10^{-3}\,\text{A}}{\exp\left(\frac{0.75\,V}{0.026\,V}\right) - 1} = 2.97 \times 10^{-16}\,\text{A} \tag{2}$$

Note that the thermal voltage (V_T) was given in this particular question.

4.2 Part B: If the diode cross-section area is now doubled, what voltage yields a current of 1 mA?

As a physical property of a specific diode, the saturation current I_S is determined by

$$I_S = A\,q\,n_i^2 \left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right) \tag{3}$$

From (3), we can see the proportional correlation between I_S and the cross-section area A. So, if A is doubled, I_S will be doubled as a result. Let this doubled saturation current be I_S':

$$I_S' = 2I_S = 5.94 \times 10^{-16}\,\text{A} \tag{4}$$

Now, we have the new saturation current I_S'. With the diode current unchanged, we can determine the new diode voltage, V_D' :

$$V_D' = V_T \ln\left(\frac{I_D}{I_S'} + 1 \right) \tag{5}$$

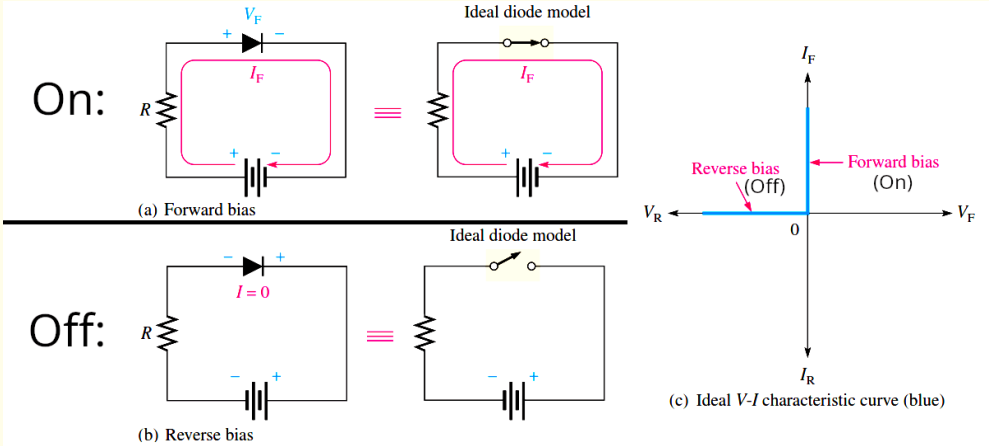
$$= (0.026\,\text{V}) \ln\left(\frac{1 \times 10^{-3}\,\text{A}}{5.94 \times 10^{-16}\,\text{A}} + 1 \right) = 0.732\ \text{V} \tag{6}$$

The Ideal Diode Model

The ideal diode model is a simplified approximation that makes diode circuit analysis much easier:

Key Assumptions:

- When forward biased ($V_D > 0$): The diode acts as a **short circuit** (closed switch)
  - $V_D = 0$ V (no voltage drop across the diode)
  - $I_D > 0$ A (current flows freely)
- When reverse biased ($V_D < 0$): The diode acts as an **open circuit** (open switch)
  - $I_D = 0$ A (no current flows)
  - $V_D < 0$ V (diode sustains reverse voltage)



The ideal diode model I-V characteristic and circuit equivalents.

5 Question 5

Find the Q-point for the diodes in the circuits in Figure 3 using the ideal diode model.

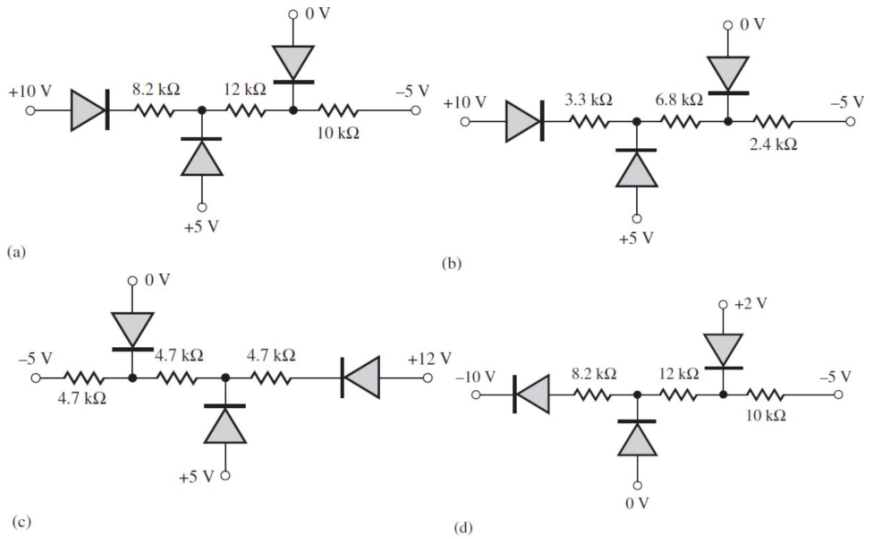


Figure 3: Q5 circuit diagram

Notes for Q5:

- The Q-point (operating point) of a diode is defined as (V_D, I_D), that is, the voltage across and current through the diode, respectively.
- To remember the polarity of the voltage (V_D) of a diode (which way is positive/negative), remember that the arrow-like symbol of the diode *"points"* to the negative side of V_D.
- For the ideal model:
  - ON state $\rightarrow$ ($V_D = 0$ V, $I_D > 0$ A)
  - OFF state $\rightarrow$ ($V_D < 0$ V, $I_D = 0$ A)

— Analysis strategy for multiple diodes —

Step 0: Label the diodes in the circuit diagram, including the direction of V_D of each (polarity).

Step 1: Make an assumption by guessing ON/OFF for each diode. Some tricks include:

- Inspect the circuit - if the diode's positive V_D is connected to the highest voltage in the circuit, it must be ON.
- If the diode's *negative* V_D is connected to the lowest voltage in the circuit (i.e. ground), it must be ON.

Step 2: Simplify the circuit

- Replace ON diodes with short circuits (i.e. wires, this sets $V_D = 0$ V).
- Replace OFF diodes with open circuits (this sets $I_D = 0$ A).

Step 3: Solve for the Q-points, which means find V_D (for OFF diodes) or find I_D (for ON diodes) for each and every diode in the circuit.

- This means you have to use circuit analysis on the simplified circuit from the previous step.
- Typically, this involves using KVL and KCL as you would on a standard circuit.

Step 4: Check the assumptions made:

- If any of the diodes that were assumed ON has $I_D < 0$, then the assumption was incorrect.
- If any of the diodes that were assumed OFF has $V_D > 0$, then the assumption was incorrect.

Repeat: If any assumptions were incorrect, start a new trial from Step 1 with corrected assumptions.

- On the next trial, flip one (or more) of the diodes that were incorrect from ON to OFF (or vice versa).

The following solutions demonstrate this strategy.

Solution:

5.1 Part A

Step 0: Label Diodes
See Figure 4, which has diodes and node voltages labelled, as referenced in the text below.

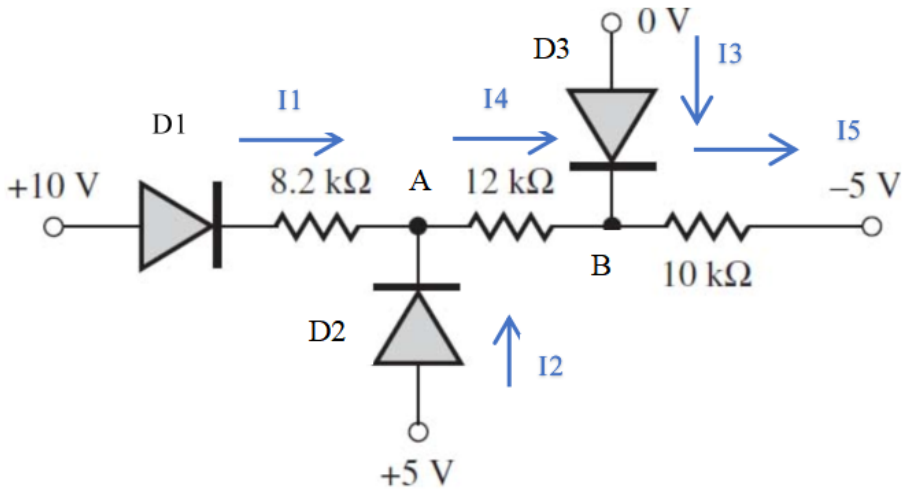


Figure 4: Part A circuit diagram

— Trial 1 —

Step 1: Make an assumption
We assume that all diodes D_1, D_2, and D_3 are ON.

Step 2: Simplify the circuit
With all diodes ON (short circuits), node A is shorted to the $+5\,\text{V}$ voltage source, and node B is shorted to the $0\,\text{V}$ voltage source (ground). These short circuits force the voltages at nodes A and B to be:

$$V_A = 5\,\text{V}, \qquad V_B = 0\,\text{V}.$$

Step 3: Solve for the Q-points
According to Ohm's law, the currents through the resistors are determined using the values $V_A = 5\,\text{V}$ and $V_B = 0\,\text{V}$ found in the simplification step:

$$I_1 = \frac{10\,\text{V} - V_A}{8.2\,\text{k}\Omega} = \frac{10\,\text{V} - 5\,\text{V}}{8.2\,\text{k}\Omega} = 0.61\,\text{mA} \tag{1}$$

$$I_4 = \frac{V_A - V_B}{12\,\text{k}\Omega} = \frac{5\,\text{V} - 0\,\text{V}}{12\,\text{k}\Omega} = 0.42\,\text{mA} \tag{2}$$

$$I_5 = \frac{V_B - (-5\,\text{V})}{10\,\text{k}\Omega} = \frac{0\,\text{V} - (-5\,\text{V})}{10\,\text{k}\Omega} = 0.5\,\text{mA} \tag{3}$$

Using KCL at node A: $I_1 + I_2 = I_4$, therefore:

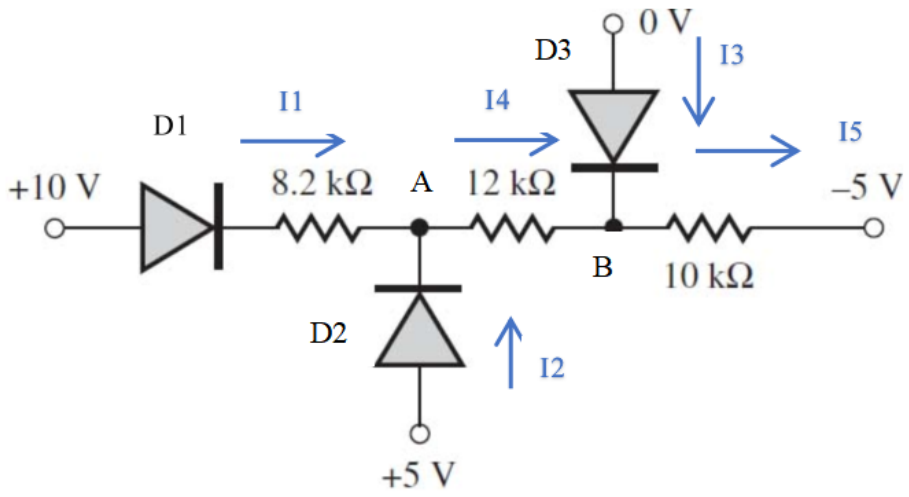
$$I_2 = I_4 - I_1 = 0.42\,\text{mA} - 0.61\,\text{mA} = -0.19\,\text{mA} < 0 \Rightarrow D_2 \text{ not ON!}$$

For D_1, we have $I_1 = 0.61\,\text{mA} > 0$, which would confirm D_1 is ON.
For D_3, using KCL at node B: $I_3 + I_4 = I_5$, therefore:

$$I_3 = I_5 - I_4 = 0.5\,\text{mA} - 0.42\,\text{mA} = 0.08\,\text{mA} > 0 \Rightarrow D_3 \text{ ON}$$

Step 4: Check the assumptions
We found that D_2 has a negative current I_2 through it, which means that D_2 cannot be ON. We proceed to try another guess with D_2 set to OFF.

— **Trial 2** —



Step 1: Make an assumption
Now, assume D_1 and D_3 are ON, D_2 is OFF.
Step 2: Simplify the circuit
With D_3 ON (short circuits), node B is shorted to the 0 V voltage (i.e. ground), so $V_B = 0\,\text{V}$. With D_2 OFF (open circuit), we have $I_2 = 0\,\text{A}$. We must determine V_A through circuit analysis.
Step 3: Solve for the Q-points
Since D_2 is OFF (open circuit) with $I_2 = 0\,\text{A}$, the 8.2 kΩ and 12 kΩ resistors are in series:

$$I_1 = I_4 = \frac{10\,\text{V} - 0\,\text{V}}{(8.2 + 12)\,\text{k}\Omega} = \frac{10\,\text{V}}{20.2\,\text{k}\Omega} = 0.495\,\text{mA} \Rightarrow D_1 \text{ ON} \tag{4}$$

$$I_5 = \frac{0\,\text{V} - (-5\,\text{V})}{10\,\text{k}\Omega} = 0.5\,\text{mA} \tag{5}$$

$$I_3 = I_5 - I_4 = 0.5\,\text{mA} - 0.495\,\text{mA} = 0.005\,\text{mA} \Rightarrow D_3 \text{ ON} \tag{6}$$

Note that because I_2 is zero, then $I_1 = I_4$ and we can also treat the 8.2 kΩ and 12 kΩ resistors as an equivalent series resistor (we cannot do this when I_2 is nonzero).

To find the voltage across D_2, we need V_A:

$$V_{D_2} = 5\,\text{V} - V_A \tag{7}$$

Using Ohm's law:

$$\begin{aligned} V_A &= 10\,\text{V} - I_1 \cdot 8.2\,\text{k}\Omega \\ &= 10\,\text{V} - 0.495\,\text{mA} \cdot 8.2\,\text{k}\Omega \\ &= 10\,\text{V} - 4.06\,\text{V} = 5.941\,\text{V} \end{aligned} \tag{8}$$

Now we can find the voltage across the OFF diode:

$$V_{D_2} = 5\,\text{V} - V_A = 5\,\text{V} - 5.941\,\text{V} = -0.941\,\text{V} < 0 \Rightarrow D_2 \text{ OFF} \tag{9}$$

Step 4: Check the assumptions
All assumptions are consistent: D_1 is ON with $I_{D_1} = 0.495\,\text{mA} > 0$, D_2 is OFF with $V_{D_2} = -0.941\,\text{V} <$

0 and D_3 is ON with $I_{D_3} = 0.005\,\text{mA} > 0$. The assumptions from Trial 2 are correct! The final Q-points are:

$$Q_{D_1} = (I_{D_1} = 0.495\,\text{mA},\ V_{D_1} = 0\,\text{V}) \Rightarrow \text{ON}$$
$$Q_{D_2} = (I_{D_2} = 0\,\text{A},\ V_{D_2} = -0.941\,\text{V}) \Rightarrow \text{OFF}$$
$$Q_{D_3} = (I_{D_3} = 0.005\,\text{mA},\ V_{D_3} = 0\,\text{V}) \Rightarrow \text{ON}$$

5.2 Part B

Step 0: Label Diodes
See Figure 5, which has diodes and node voltages labelled, as referenced in the text below.

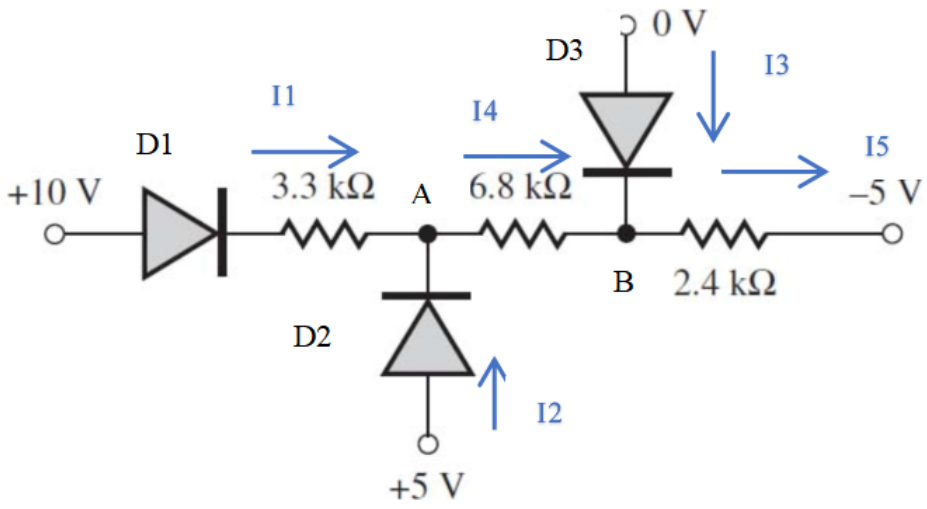


Figure 5: Part B circuit diagram

— **Trial 1** —

Step 1: Make an assumption
We assume that all diodes D_1, D_2, and D_3 are ON.
Step 2: Simplify the circuit
With all diodes ON (short circuits), node A is shorted to the $+5\,\text{V}$ voltage source, and node B is shorted to the $0\,\text{V}$ voltage (ground). These short circuits force the voltages at nodes A and B to be:

$$V_A = 5\,\text{V}, \qquad V_B = 0\,\text{V}.$$

Step 3: Solve for the Q-points
According to Ohm's law, the currents through the resistors are determined using the values $V_A = 5\,\text{V}$ and $V_B = 0\,\text{V}$ found in the simplification step:

$$I_1 = \frac{10\,\text{V} - V_A}{3.3\,\text{k}\Omega} = \frac{10\,\text{V} - 5\,\text{V}}{3.3\,\text{k}\Omega} = 1.52\,\text{mA} \Rightarrow D_1 \text{ ON} \tag{10}$$

$$I_4 = \frac{V_A - V_B}{6.8\,\text{k}\Omega} = \frac{5\,\text{V} - 0\,\text{V}}{6.8\,\text{k}\Omega} = 0.74\,\text{mA} \tag{11}$$

$$I_5 = \frac{V_B - (-5\,\text{V})}{2.4\,\text{k}\Omega} = \frac{0\,\text{V} - (-5\,\text{V})}{2.4\,\text{k}\Omega} = 2.08\,\text{mA} \tag{12}$$

Using KCL at node B: $I_3 + I_4 = I_5$, therefore:

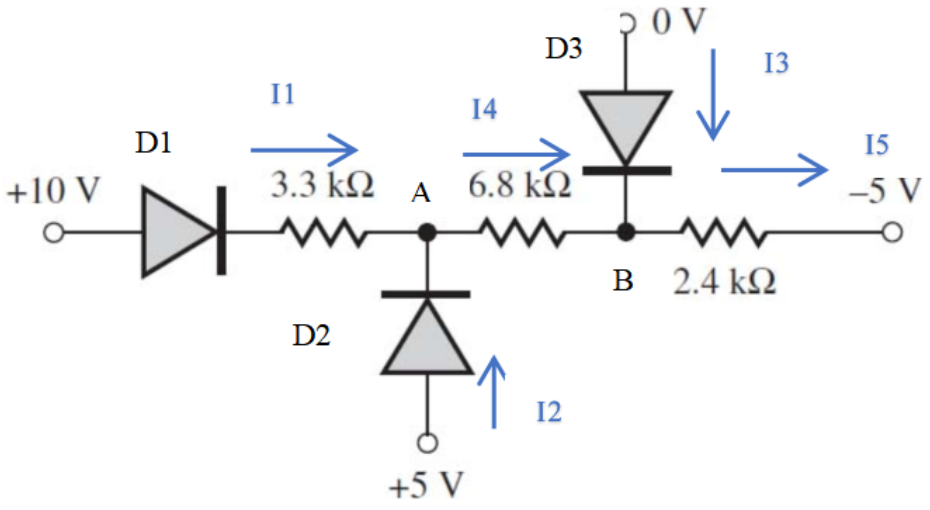
$$I_3 = I_5 - I_4 = 2.08\,\text{mA} - 0.74\,\text{mA} = 1.34\,\text{mA} > 0 \Rightarrow D_3 \text{ ON}$$

Using KCL at node A: $I_1 + I_2 = I_4$, therefore:

$$I_2 = I_4 - I_1 = 0.74\,\text{mA} - 1.52\,\text{mA} = -0.78\,\text{mA} < 0 \Rightarrow D_2 \text{ not ON!}$$

Step 4: Check the assumptions
We found that D_2 has a negative current I_2 through it, which means that D_2 cannot be ON. We proceed to try another guess with D_2 set to OFF:

— Trial 2 —



Step 1: Make an assumption
Now, assume D_1 and D_3 are ON, while D_2 is OFF.
Step 2: Simplify the circuit
With D_1 and D_3 ON (short circuits), node A is shorted to the $+5\,\text{V}$ voltage source, so $V_A = 5\,\text{V}$, and node B is shorted to the $0\,\text{V}$ voltage (ground), so $V_B = 0\,\text{V}$. With D_2 OFF (open circuit), we have $I_2 = 0\,\text{A}$.
Step 3: Solve for the Q-points
Since D_2 is OFF (open circuit) with $I_2 = 0\,\text{A}$, the $3.3\,\text{k}\Omega$ and $6.8\,\text{k}\Omega$ resistors are in series:

$$I_1 = I_4 = \frac{10\,\text{V} - 0\,\text{V}}{(3.3 + 6.8)\,\text{k}\Omega} = \frac{10\,\text{V}}{10.1\,\text{k}\Omega} = 0.99\,\text{mA} \Rightarrow D_1 \text{ ON} \tag{13}$$

$$I_5 = \frac{0\,\text{V} - (-5\,\text{V})}{2.4\,\text{k}\Omega} = 2.08\,\text{mA} \tag{14}$$

$$I_3 = I_5 - I_4 = 2.08\,\text{mA} - 0.99\,\text{mA} = 1.09\,\text{mA} \Rightarrow D_3 \text{ ON} \tag{15}$$

Note that we can treat the $3.3\,\text{k}\Omega$ and $6.8\,\text{k}\Omega$ resistors as in series because I_2 is zero (we cannot do this when I_2 is nonzero).
To find the voltage across D_2, we need V_A. Using Ohm's law:

$$\begin{aligned} V_A &= 10\,\text{V} - I_1 \cdot 3.3\,\text{k}\Omega \\ &= 10\,\text{V} - 0.99\,\text{mA} \cdot 3.3\,\text{k}\Omega \\ &= 10\,\text{V} - 3.27\,\text{V} = 6.73\,\text{V} \end{aligned} \tag{16}$$

Now we can find the voltage across the OFF diode:

$$V_{D_2} = 5\,\text{V} - V_A = 5\,\text{V} - 6.73\,\text{V} = -1.73\,\text{V} < 0 \Rightarrow D_2 \text{ OFF} \tag{17}$$

Step 4: Check the assumptions
All assumptions are consistent: D_1 is ON with $I_{D_1} = 0.99\,\text{mA} > 0$, D_2 is OFF with $V_{D_2} = -1.73\,\text{V} < 0$ and D_3 is ON with $I_{D_3} = 1.09\,\text{mA} > 0$. The assumptions from Trial 2 are correct! The final Q-points are:

$$\begin{aligned} Q_{D_1} &= (I_{D_1} = 0.99\,\text{mA},\ V_{D_1} = 0\,\text{V}) \Rightarrow \text{ON} \\ Q_{D_2} &= (I_{D_2} = 0\,\text{A},\ V_{D_2} = -1.73\,\text{V}) \Rightarrow \text{OFF} \\ Q_{D_3} &= (I_{D_3} = 1.09\,\text{mA},\ V_{D_3} = 0\,\text{V}) \Rightarrow \text{ON} \end{aligned}$$

5.3 Part C

Step 0: Label Diodes
See Figure 6, which has diodes and node voltages labelled, as referenced in the text below.

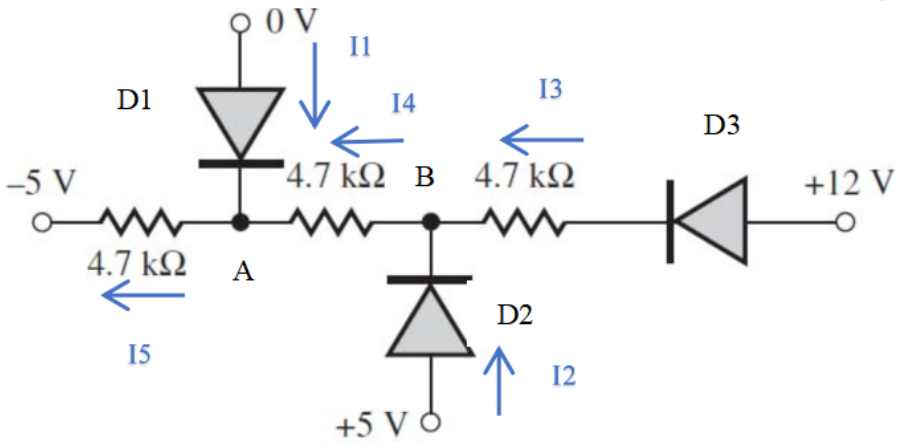


Figure 6: Part C circuit diagram

— **Trial 1** —

Step 1: Make an assumption
We assume that all diodes D_1, D_2, and D_3 are ON.
Step 2: Simplify the circuit
With all diodes ON (short circuits), node A is shorted to the 0 V voltage (ground), and node B is shorted to the +5 V voltage source. These short circuits force the voltages at nodes A and B to be:

$$V_A = 0\,\text{V}, \qquad V_B = 5\,\text{V}.$$

Step 3: Solve for the Q-points
According to Ohm's law, the three currents through the resistors are determined using the values $V_A = 0\,\text{V}$ and $V_B = 5\,\text{V}$ found in the simplification step:

$$I_3 = \frac{12\,\text{V} - V_B}{4.7\,\text{k}\Omega} = \frac{12\,\text{V} - 5\,\text{V}}{4.7\,\text{k}\Omega} = 1.49\,\text{mA} \tag{18}$$

$$I_4 = \frac{V_B - V_A}{4.7\,\text{k}\Omega} = \frac{5\,\text{V} - 0\,\text{V}}{4.7\,\text{k}\Omega} = 1.06\,\text{mA} \tag{19}$$

$$I_5 = \frac{V_A - (-5\,\text{V})}{4.7\,\text{k}\Omega} = \frac{0\,\text{V} - (-5\,\text{V})}{4.7\,\text{k}\Omega} = 1.06\,\text{mA} \tag{20}$$

Using KCL at node A: $I_1 + I_4 = I_5$, therefore:

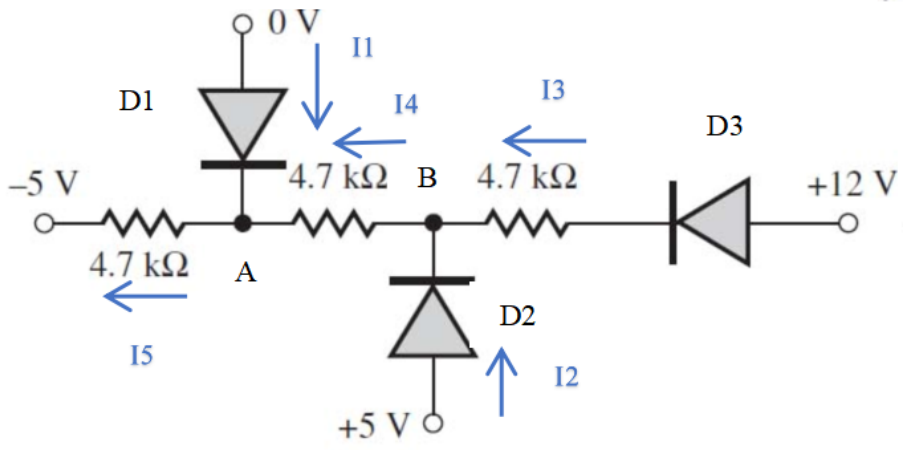
$$I_1 = I_5 - I_4 = 1.06\,\text{mA} - 1.06\,\text{mA} = 0\,\text{mA} \Rightarrow D_1 \text{ not ON!}$$

Using KCL at node B: $I_2 + I_3 = I_4$, therefore:

$$I_2 = I_4 - I_3 = 1.06\,\text{mA} - 1.49\,\text{mA} = -0.43\,\text{mA} < 0 \Rightarrow D_2 \text{ not ON!}$$

For D_3, we have $I_3 = 1.49\,\text{mA} > 0$, which confirms D_3 is ON.
Step 4: Check the assumptions
We found that D_1 has zero current (borderline case) and D_2 has negative current. Since we assumed both were ON, this is impossible. We need to try another guess with D_1 and D_2 set to OFF.

— **Trial 2** —



Step 1: Make an assumption
Now, assume D_1 and D_2 are OFF, while D_3 is ON.
Step 2: Simplify the circuit
With D_1 and D_2 OFF (open circuits), we have $I_1 = 0\,\text{A}$ and $I_2 = 0\,\text{A}$. With D_3 ON (short circuit), we have $V_{D_3} = 0\,\text{V}$. We must determine both V_A and V_B through circuit analysis.
Step 3: Solve for the Q-points
Since D_1 and D_2 are OFF (open circuits) and D_3 is ON, all the current must flow through the series path from 12 V through the three 4.7 kΩ resistors to −5 V:

$$\begin{aligned} I_5 = I_4 = I_3 &= \frac{12\,\text{V} - (-5\,\text{V})}{4.7\,\text{k}\Omega + 4.7\,\text{k}\Omega + 4.7\,\text{k}\Omega} \\ &= \frac{17\,\text{V}}{14.1\,\text{k}\Omega} = 1.21\,\text{mA} \Rightarrow D_3 \text{ ON} \end{aligned} \tag{21}$$

To find the voltages across D_1 and D_2, we need to determine V_A and V_B. Using Ohm's law from the left (rearranged for V_A):

$$\begin{aligned} V_A &= -5\,\text{V} + I_5 \cdot 4.7\,\text{k}\Omega \\ &= -5\,\text{V} + 1.21\,\text{mA} \cdot 4.7\,\text{k}\Omega \\ &= -5\,\text{V} + 5.67\,\text{V} = 0.67\,\text{V} \end{aligned} \tag{22}$$

And from the right (rearranged for V_B):

$$\begin{aligned} V_B &= 12\,\text{V} - I_3 \cdot 4.7\,\text{k}\Omega \\ &= 12\,\text{V} - 1.21\,\text{mA} \cdot 4.7\,\text{k}\Omega \\ &= 12\,\text{V} - 5.67\,\text{V} = 6.33\,\text{V} \end{aligned} \tag{23}$$

Now we can find the voltages across the OFF diodes:

$$V_{D_1} = 0\,\text{V} - V_A = 0\,\text{V} - 0.67\,\text{V} = -0.67\,\text{V} < 0 \Rightarrow D_1 \text{ OFF} \tag{24}$$
$$V_{D_2} = 5\,\text{V} - V_B = 5\,\text{V} - 6.33\,\text{V} = -1.33\,\text{V} < 0 \Rightarrow D_2 \text{ OFF} \tag{25}$$

Step 4: Check the assumptions
All assumptions are consistent: D_1 is OFF with $V_{D_1} = -0.67\,\text{V} < 0$, D_2 is OFF with $V_{D_2} = -1.33\,\text{V} < 0$ and D_3 is ON with $I_{D_3} = 1.21\,\text{mA} > 0$. The final Q-points are:

$$\begin{aligned} Q_{D_1} &= (I_{D_1} = 0\,\text{A},\ V_{D_1} = -0.67\,\text{V}) \Rightarrow \text{OFF} \\ Q_{D_2} &= (I_{D_2} = 0\,\text{A},\ V_{D_2} = -1.33\,\text{V}) \Rightarrow \text{OFF} \\ Q_{D_3} &= (I_{D_3} = 1.21\,\text{mA},\ V_{D_3} = 0\,\text{V}) \Rightarrow \text{ON} \end{aligned}$$

5.4 Part D

Step 0: Label Diodes
See Figure 7, which has diodes and node voltages labelled, as referenced in the text below.

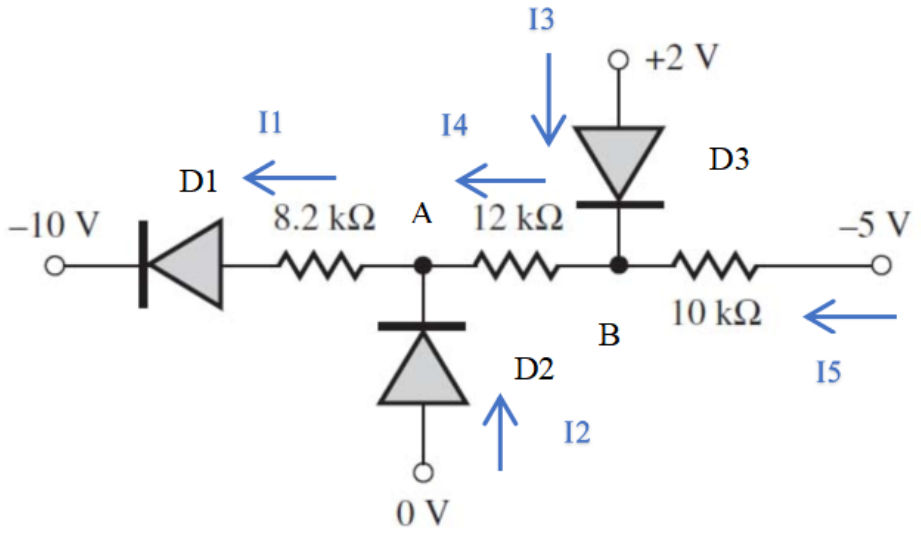


Figure 7: Part D circuit diagram

— Trial 1 —

Step 1: Make an assumption
We assume that all diodes D_1, D_2, and D_3 are ON.
Step 2: Simplify the circuit
With all diodes ON (short circuits), node A is shorted to the 0 V voltage (ground), and node B is shorted to the 2 V voltage source. These short circuits force the voltages at nodes A and B to be:

$$V_A = 0\,\text{V}, \qquad V_B = 2\,\text{V}.$$

Step 3: Solve for the Q-points
According to Ohm's law, the currents through the resistors are determined using the values $V_A = 0\,\text{V}$ and $V_B = 2\,\text{V}$ found in the simplification step:

$$I_1 = \frac{V_A - (-10\,\text{V})}{8.2\,\text{k}\Omega} = \frac{0\,\text{V} - (-10\,\text{V})}{8.2\,\text{k}\Omega} = 1.220\,\text{mA} \Rightarrow D_1 \text{ ON} \tag{26}$$

$$I_4 = \frac{V_B - V_A}{12\,\text{k}\Omega} = \frac{2\,\text{V} - 0\,\text{V}}{12\,\text{k}\Omega} = 0.167\,\text{mA} \tag{27}$$

$$I_5 = \frac{(-5\,\text{V}) - V_B}{10\,\text{k}\Omega} = \frac{(-5\,\text{V}) - 2\,\text{V}}{10\,\text{k}\Omega} = -0.7\,\text{mA} \tag{28}$$

Note that I_5 represents the current flowing from right to left through the 10 kΩ resistor. Since we got a negative value, the current actually flows from left to right (from node B toward $-5\,\text{V}$).
Using KCL at node A: $I_2 + I_4 = I_1$, therefore:

$$I_2 = I_1 - I_4 = 1.220\,\text{mA} - 0.167\,\text{mA} = 1.053\,\text{mA} > 0 \Rightarrow D_2 \text{ ON}$$

Using KCL at node B: $I_4 = I_3 + I_5$, therefore:

$$I_3 = I_4 - I_5 = 0.167\,\text{mA} - (-0.7\,\text{mA}) = 0.867\,\text{mA} > 0 \Rightarrow D_3 \text{ ON}$$

Step 4: Check the assumptions
All three diode currents are positive: $I_{D_1} = 1.220\,\text{mA} > 0$, $I_{D_2} = 1.053\,\text{mA} > 0$ and $I_{D_3} = 0.867\,\text{mA} >$

0. All assumptions are consistent. This is a case where all three diodes are indeed ON in the final solution. The final Q-points are:

$$Q_{D_1} = (I_{D_1} = 1.22\,\text{mA},\ V_{D_1} = 0\,\text{V}) \Rightarrow \text{ON}$$
$$Q_{D_2} = (I_{D_2} = 1.05\,\text{mA},\ V_{D_2} = 0\,\text{V}) \Rightarrow \text{ON}$$
$$Q_{D_3} = (I_{D_3} = 0.867\,\text{mA},\ V_{D_3} = 0\,\text{V}) \Rightarrow \text{ON}$$